Ogbonna Arinze

Berlin, Germany | +49 152 35797306 | arinzeogbo@gmail.com | LinkedIn | arinze.online

SUMMARY

Senior Fullstack Engineer with 8 years of end-to-end **TypeScript** delivery across regulated industries, including **healthcare** at Genentech (Roche), **fintech** at ActivEdge banks, and government data transparency at OurNigeria. Built a Retrieval-Augmented Generation chat widget at Genentech, then took it further at OurNigeria with a **multi-agent AI system** on the Mastra framework, hybrid RAG pipelines, and conversational workflow execution. Strong product instinct for compliance-sensitive UX (audit trails, RBAC, JWT auth) plus deep **Node.js** backend experience with NestJS, Prisma, Postgres, and MongoDB.

CORE COMPETENCIES

- **End-to-end TypeScript:** NestJS and Node.js on the backend, React and Next.js on the frontend, Prisma + PostgreSQL and MongoDB on the data layer.
- **Workflow and Agentic AI:** Mastra multi-agent framework, OpenAI API, LangChain, Vercel AI SDK, RAG pipelines (pgvector, BM25, RRF, Cohere rerank), prompt engineering, intent classification.
- **Healthcare/Regulated UX:** Genentech clinical-trial site integrations, JWT auth with HTTP-only cookies, role-based access control, audit-friendly migrations.
- **Performance and Reliability:** Lighthouse profiling, code splitting, bundle analysis, Docker Compose with health checks, memory limits, and graceful shutdown.
- **API Design:** REST APIs, GraphQL, SSE streaming with keepalive heartbeats, webhook integrations, JWT.
- **DevOps and Tooling:** Docker, GitHub Actions CI/CD, Nx and pnpm monorepos, Infisical secrets, Langfuse observability.
- **Leadership:** Cross-functional product partnership with designers and PMs, mentorship of juniors and interns.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **Retrieval-Augmented Generation** system using OpenAI GPT models, **LangChain**, **Vercel AI SDK**, and **Chroma**, letting users connect data sources, embed the widget on external sites, and query their data conversationally.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** SPA to a **React** Chrome extension with **TypeScript** for site-to-EHR integrations, owning the rebuild end-to-end.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend with HTTP-only cookies, protected routes, permission-gated UI, and seamless token refresh, partnered with backend on the contract.
- Architected a reusable design system in **React** and **Radix UI** with **Tailwind CSS**, raising UI consistency, accessibility, and developer velocity.
- Introduced **Redux** for client state and **React Query** for server-state caching and request deduplication, refactored class components to functional.
- Optimized application performance through lazy loading, code splitting, image optimization, memoization, and CDN delivery, profiled with **Lighthouse** and Chrome DevTools.
- Integrated **Playwright** E2E into **GitHub Actions** CI/CD with **Allure Reports** on GitHub Pages and Slack notifications on every push.
- Maintained the existing **Vue.js** codebase in parallel with the React rebuild, fixing bugs and shipping features without regression.
- Mentored two junior developers (interns) through written-first reviews and pair programming.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of a greenfield **React** + **TypeScript** cloud console interface, achieving a **25% boost in user engagement** and accelerating delivery by **30%**.
- Profiled render bottlenecks with **React Profiler** and **Lighthouse**, cut bundle size **30-40%**, achieving **35% faster** initial load.
- Drove a **40% improvement** in runtime performance by reworking caching strategies.
- Introduced **Jest** and **Enzyme** testing plus **Storybook**, leading to a **20% drop** in bug reports.
- Established team engineering standards via **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions**.
- Mentored junior engineers on frontend best practices.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of transactions daily and boosting settlement rates by **80%**.

- Developed payment modules (MasterCard, VISA, Express-cash, E-commerce) with **React**, **Redux**, **RxJS**, and **Axios**.
- Built a real-time account reconciliation tool, lifting daily reconciliation by **60%** via new transaction processing strategies.
- Adopted the native **EventSource API** over socket.io for real-time data streaming.
- Built dashboards with **react-chartjs**, **Redux Toolkit**, and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into efficient **MongoDB** schemas and developed **NodeJS** RESTful APIs.
- Converted Figma designs into reusable **React** components.
- Engineered a CI/CD pipeline for automated API deployment to Heroku and AWS production.
- Mentored two interns through frontend and backend project work.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Built **SureEdu**, a school management portal, in a team of 5: staff payroll/attendance, student records, guardian portals.
- Designed **MongoDB** schemas and implemented server-side **NodeJS** logic with REST endpoints.
- Authored use cases, translated them into database models, and wrote feature documentation per application.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget transparency platform with workflow-style multi-agent execution over 700+ documents and 708K+ vector embeddings.

- Architected the platform as an **Nx monorepo** of 6 apps: **NestJS** API, **NestJS** ingestion pipeline, **Next.js** citizen web, admin dashboard, marketing landing, Remotion video service, orchestrated with **pnpm**.
- Designed and built a **multi-agent AI system** on the **Mastra** framework with 5 specialized agents (Router, Budget, Corruption, GovSpend, Impact) using three-stage intent classification (keyword scoring, LLM classification with 5-min cache, inferred fallback) and domain-specific RAG analysis - the agentic-workflow shape that maps directly to a no-code automation Copilot.
- Built a hybrid **RAG** pipeline (pgvector + BM25 + Reciprocal Rank Fusion + Cohere rerank) over 708K+ Voyage AI embeddings (1024-dim) across 4 vector indexes.
- Implemented **SSE streaming** chat with keepalive heartbeats, token-budgeted context windows, conversation summarization, and memory extraction; phone OTP and Telegram OAuth auth with session cookies.
- Designed the **PostgreSQL 16** schema with **Prisma v7** and pgvector, with custom migration tooling that filters Mastra-managed chunk tables out of Prisma drift detection.
- Configured **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown; integrated **Langfuse** for LLM observability and **Infisical** for secrets management.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*